

Task 4 – Beamforming Simulator

Beamforming, phased arrays, and direction of arrival (DOA) are fundamental concepts in modern technologies starting from wireless communications, 5G, radar, sonar to biomedical applications like ultrasound and tumor ablations. The core ideas of beamforming are delays, phase-shifts and constructive/destructive interference, which are all direct applications for Fourier transform properties. Do your own research and/or check the tutorials below ([tutorial 1](#) – check all videos in the list, [tutorial 2](#), [tutorial 3](#)) to grasp the idea and inspired by the relevant available toolboxes (e.g., Matlab's and others) develop your own 2D beamforming simulator where the user can:

- Customize the beamforming and oscillators main parameters (7 parameters at least) below to be able to control the output waveform, and steer the beam direction in real time.
- Visualize the constructive/destructive map along with the beam profile in different synchronized viewers.
- Allow the user to reduce the side lobes of any beamforming profile you generate. Do your research on how to it by searching for “apodization” and “windowing”, and include this utility in your simulator.
- Assume a specific SNR for your beams. This should reflect on any output measures in all the below applications. The user can customize this applied SNR from 0 to 1000.

Then, based on your simulator, provide the three following applications:

1. **5G simulator:** where the user can place 3 5G towers, and two network users that can be moved via keyboard shortcut. Your system should:
 - a. Highlight the beam connectivity between each tower and each user based on the distance between each pair.
 - b. Automatically update each tower customizable parameters (these updates should be clearly visible to the user) to reflect the proper beamforming.
 - c. Support that one tower can communicate with multiple users if they both exist in its influence area (check the video [here](#) for some hints).
2. **Ultrasound simulator:** where the user can simulate an A-, B-, and Doppler modes of ultrasound scan on a Shepp–Logan phantom.
 - a. Assume the necessary ultrasound properties for each shape inside the phantom to provide a realistic A-mode output. The user can 1) hover on any of these shapes to see its parameters, and 2) click on any of them to edit these parameters. Please, do your research on the US modes to determine which properties should be provided for the user for such purpose.
 - b. The ultrasound probe can only be put on the phantom outside surface but the user can customize its beam direction to scan the desired direction and visualize its A-mode output.
 - c. By moving the probe across the surface and in different direction, you can build a simple B-mode output.
 - d. Assume some blood vessel with some moving blood velocity and direction (you will need to simulate this on your own since it is not in the phantom by default), use the return of your probe signal to provide the Doppler mode output. The user should be able to customize the vessel direction and blood velocity.
3. **Radar simulator:** where the phased array oscillators can work in the full 360° to provide a synchronized beam that rotates in 360° in the 2D plan (rotation is done through steering, not mechanical rotation for the source – steering is much faster than mechanical rotations), and spot any solid body encountered. The user can freely throw up to 5 solid bodies in different distances and angles around the beam source. The user can freely change the size of any of these bodies. He can also move or delete any of them bodies to study the system output changes accordingly. The user can change the scanning speed as well as the width of the scanning beam. For ex, a wider beam will do a faster scan, but to determine the body size, the beam has to be narrowed to obtain more accurate estimation for the size. Think of it as quick broad scan to detect if anybody is there, and then focus with thinner beams to detect the body size. Simulate these described scenarios.

Note: Watch out for the ranges of distances and spatial resolution, and therefore beamforming frequencies used in each application. For example, while the distances and resolutions in the 5G applications range in meters, the same distances and resolutions in the US application range in cm and mm respectively.